Question 23

How many fractions from the list below have decimal representation that terminate?

$$\frac{1}{8}, \frac{1}{9}, \frac{1}{10}, \frac{1}{11}, \ldots, \frac{1}{64}, \frac{1}{65}, \frac{1}{66}$$

Question 24

A car is travelling on a road next to the train track at constant speed of 22 km/h. A train is 250 metres long and it travels at a constant speed of 17 km/h. From the moment the car passes the rear of the train, it takes the car M minutes to reach the front. Find M.

Question 25

The absolute value of N, symbolized by $|N|$, represents the distance from the signed number N to the origin, without regard to sign. For example: $|+5| = 5$ and $|-3| = 3$. Find the sum of the integer values of N that satisfy the inequality $2 < |N - 3| < 5$.

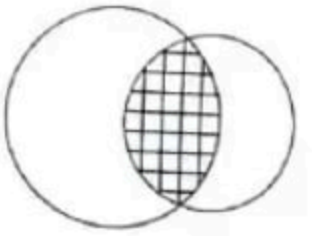
Question 22

Some 3-digit whole numbers have repeated digits. Two examples are 122 and 777.

Exactly K% of all 3-digit whole numbers have repeated digits, where K is a whole number. Find K.

Question 18

A circle of radius 14 cm and a circle of radius 21 cm overlap in the crosshatched region as shown. The area of the crosshatched region is 580 cm^2. What is the half of the total area of the two regions without any crosshatching. (Use $\pi = \frac{22}{7}$)



Question 19

Betsy collects sports posters. Exactly one of the statements below is false.

Betsy has at least 6 sports posters.

Betsy has at least 15 sports posters.

Betsy has at least 19 sports posters.

Betsy has at least 23 sports posters.

What is the greatest possible number of posters in Betsy's collection?

Question 20

XYX and YXY represent two 3-digit integers in which X and Y are distinct non-zero digits. How many different values are possible for the sum XYX + YXY?

Question 21

Find the greatest number that divides 266, 516 and 741 with same remainder in each case.

Question 17

In the 5-digit integer XYYYX, X and Y represent different non-zero digits. If XYYYX is divisible by 11, how many different values of XYYYX are possible?

Question 12

Compute the value of the expression below.

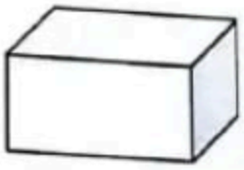
$$\frac{2019^2 - 2000^2 - 19^2}{2000 \times 19}$$

Question 13

What is the greatest number of consecutive integers that sum to -11?

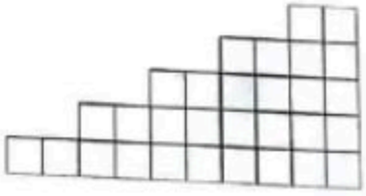
Question 14

Three of the faces of a rectangular box have areas of 30 cm^2, 18 cm^2 and 15 cm^2. What is the volume (in cm^3) of the box?



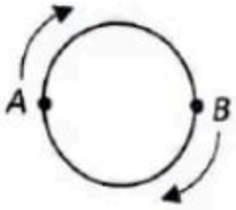
Question 15

Thirty unit squares are arranged, sharing common sides and vertices, in the "step" arrangement as shown. How many squares with integer length sides can be found in this arrangement?



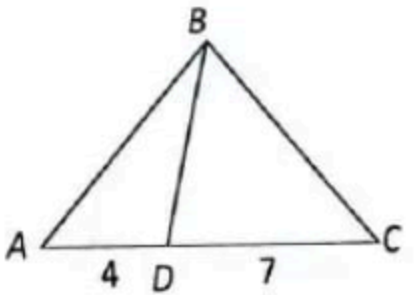
Question 16

Andy and Brett begin at opposite points on a 300-foot circular track, and jog around the track in the directions shown. Andy jogs at 4.3 feet/sec and Brett jogs at 5.5 feet/sec. How many seconds will it take for Brett and Andy to meet for the first time?



Question 10

Given ΔABC with point D on side AC such that $AD = 4$ and $CD = 7$. If the area of ΔABD is 20 square units, find the number of square units in the area of ΔABC.



Question 11

How many points (x, y) on the line segment AB with $A(-6, 5)$ and $B(10, -7)$ have integer values for both coordinates x and y?

Question 6

Let M be the greatest numeric value of $\frac{x-y}{x}$ given that x is a positive integer factor of 20 and y is a positive integer factor of 12. Find the value of 20×M.

Question 7

Let N be the greatest possible negative quotient when two distinct integers chosen from the set {−7, −5, −4, −3, −2, 0, 1, 2, 6} are divided. Find the value of $(213 + 245N)$.

Question 8

A pair of prime numbers, P and Q, have the relationship that $3P + 5Q = 121$. How many possible values of $P + Q$ are there?

Question 9

Starting at midnight, Aarf the Dog barks for 15 seconds and then is silent for the next 25 seconds. Aarf the Dog continues this bark-fest until 1:01 AM that same morning. How many times did Aarf the Dog bark for 15 seconds?

Question 5

Simplify:

$$\frac{(1+2-3)+(4+5-6)+(7+8-9)+\cdots+(97+98-99)}{3}$$

Question 1

Compute the simplified numeric value of $(16 \div 8 \times 4)$ subtracted from $(16 \times 8 \div 4)$.

Question 2

Compute the whole number value of

$$\frac{1}{3} \times \left(\frac{7981}{0.7981} - 7981\right).$$

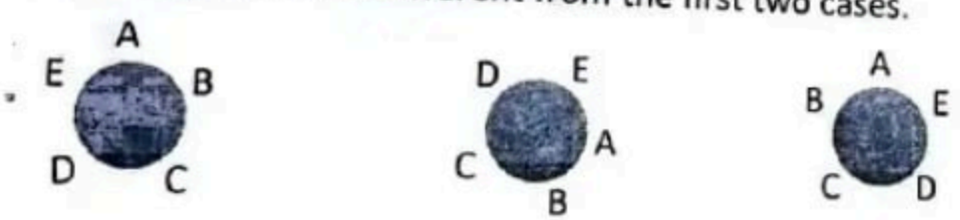
Question 3

XYX and YXY represent two 3-digit whole numbers in which X and Y are distinct non-zero digits. Find the sum of the digits of the greatest possible sum of XYX and YXY.

Question 4

Mr. and Mrs. Hansen and their three children are to be seated about a circular table. In how many different ways can the family be seated if Mr. and Mrs. Hansen are seated next to one another?

Case 1 and 2 are not considered different (one is a rotation of the other). Case 3 is different from the first two cases.



Case 1 Case 2 Case 3